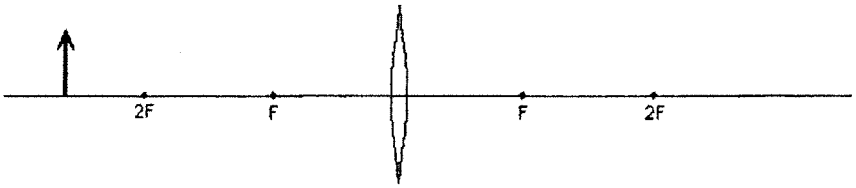
Geometric Optics

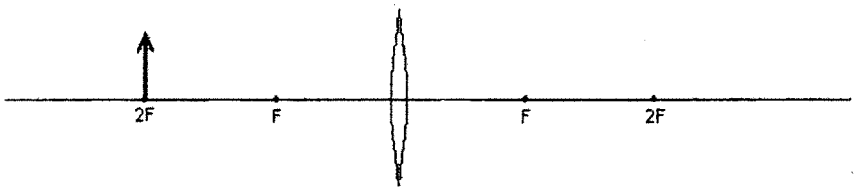
Ray Tracing WS ___

Name _____

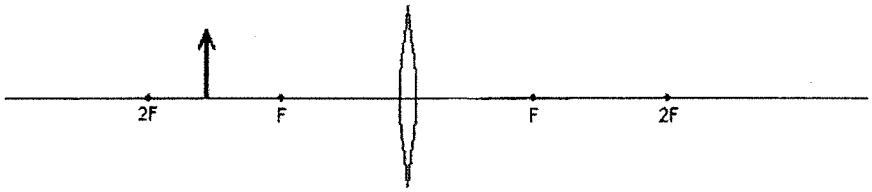
Date _____ **Per** _____



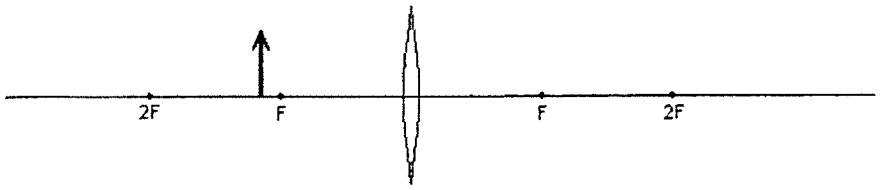
f	+	2
d_o	+	5
d_i		
h_o	+	2
h_i		
M		
Converging		
Real		
Convex		



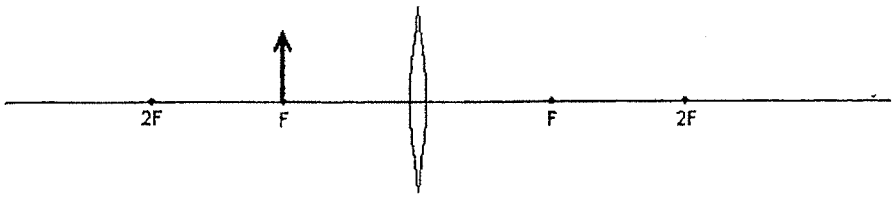
f	+	2
d_o		
d_i		
h_o	+	2
h_i		
M		



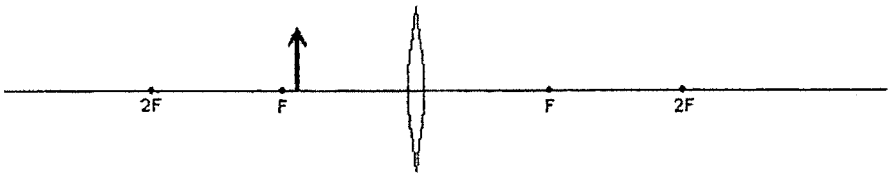
f	+	2
d_o	+	3
d_i		
h_o	+	2
h_i		
M		



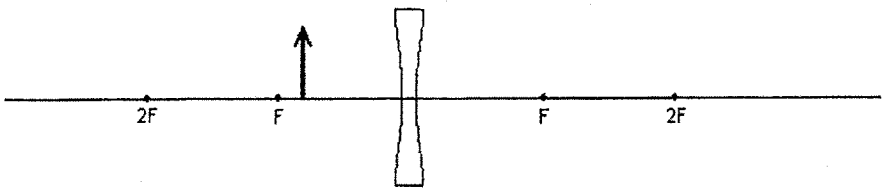
f	+	2
d_o	+	2.2
d_i		
h_o	+	2
h_i		
M		



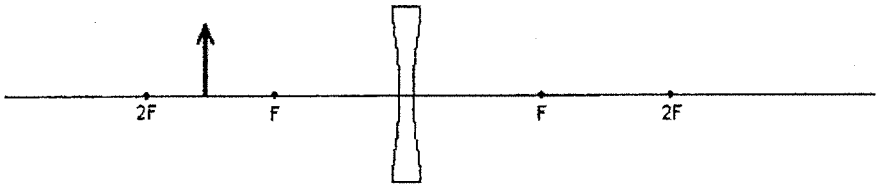
f	+	2
d_o		
d_i		
h_o	+	2
h_i		
M		



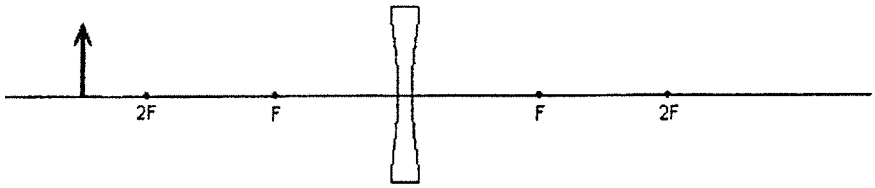
f	+	2
d_o	+	1.8
d_i		
h_o	+	2
h_i		
M		



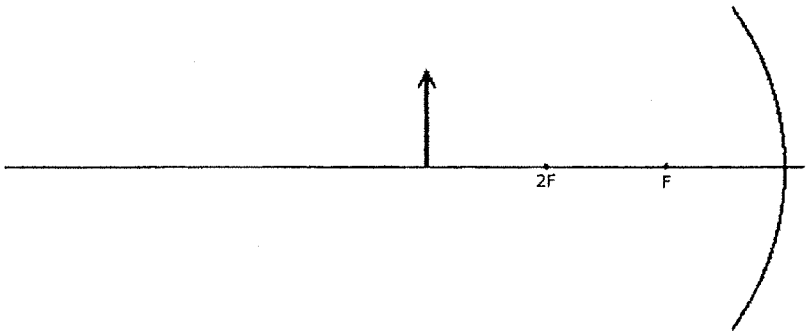
f	-	2
d_o	+	1.8
d_i		
h_o	+	2
h_i		
M		



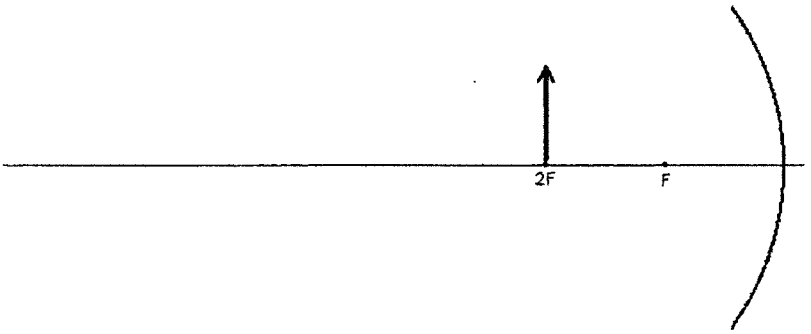
f	-	2
d_o	+	3
d_i		
h_o	+	2
h_i		
M		



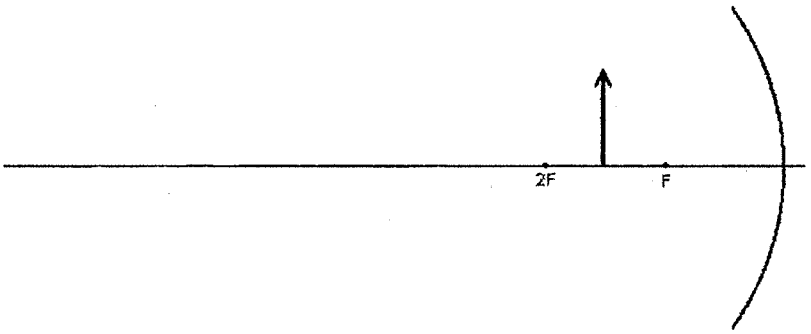
f	-	2
d_o	+	5
d_i		
h_o	+	2
h_i		
M		



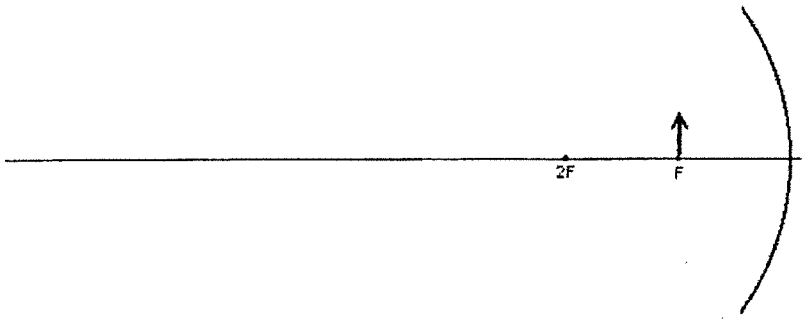
f	+	2
d_o	+	6
d_i		
h_o	+	3
h_i		
M		



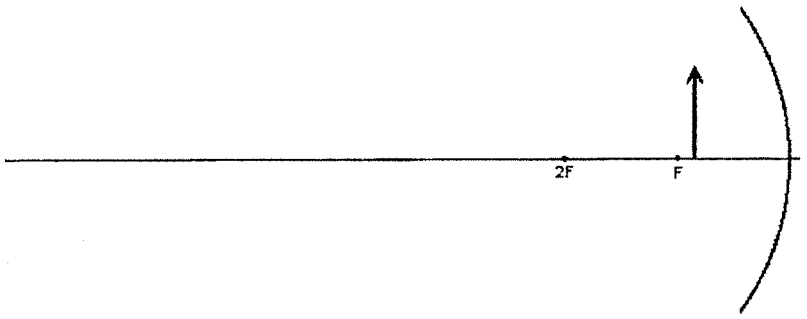
f	+	2
d_o		
d_i		
h_o	+	3
h_i		
M		



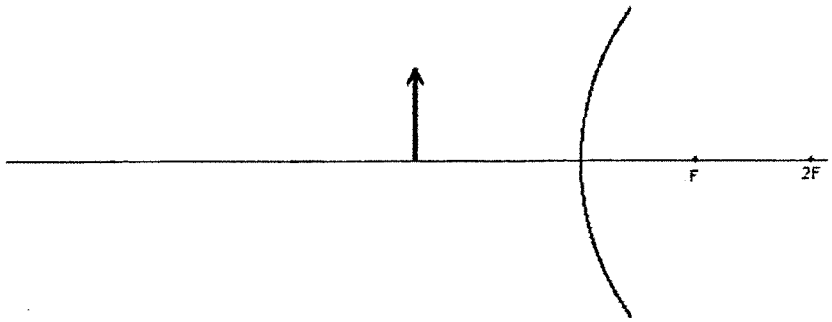
f	+	2
d_o	+	3
d_i		
h_o	+	3
h_i		
M		



f	+	2
d_o	+	2
d_i		
h_o	+	3
h_i		
M		



f	+	2
d_o	+	1.8
d_i		
h_o	+	3
h_i		
M		



f	-	2
d_o	+	3
d_i		
h_o	+	3
h_i		
M		

Geometric Optics

Name _____

Ray Tracing WS ___

Date _____ Per _____

1. Complete the tables below with the correct signs, + or – , and choose the correct terms.

	f	s_o	s_i	h_o	h_i	M	Converging or Diverging	Real or Virtual	Convex or Concave
Converging Lens, Object outside f									
Converging Lens, Object at f									
Converging Lens, Object inside f									
Converging Mirror, Object outside f									
Converging Mirror, Object at f									
Converging Mirror, Object inside f									
Diverging Lens, Object outside f									
Diverging Lens, Object inside f									
Diverging Mirror, Object outside f									
Diverging Mirror, Object inside f									

2. Which variables have the same sign in all cases above?

3. What does the sign of the focal point depend on?

4. When an image is real, what else can be said about it? (There are quite a few)

5. When an image is virtual, what else can be said about it? (There are quite a few)

6. What are the main differences between converging and diverging?

7. What are the differences between lenses and mirrors?